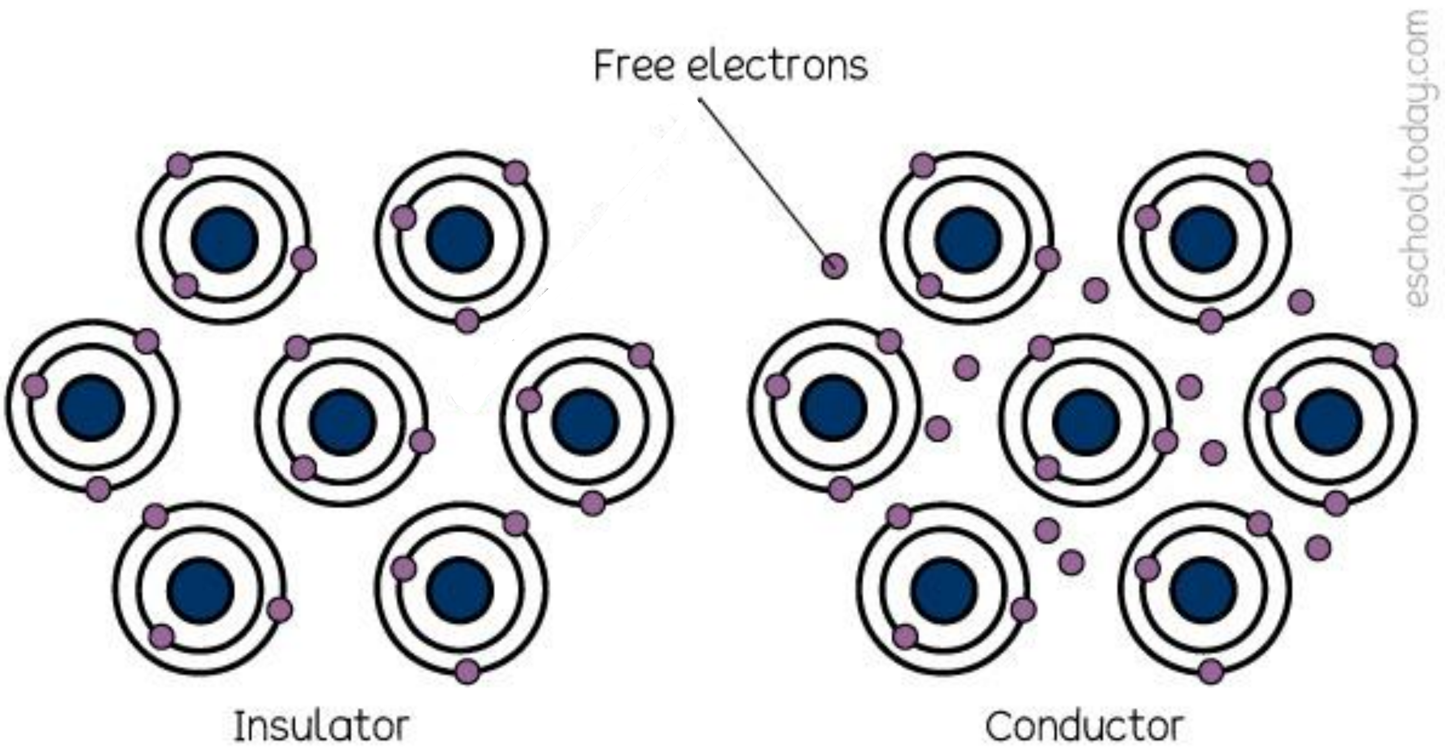
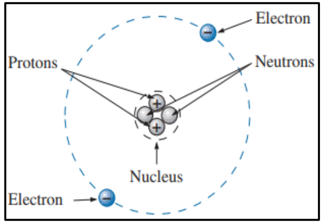

Free electrons
Insulator
Conductor
eschooltoday.com

Charge

- *Charge* is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).
- All matter is made of fundamental building blocks known as atoms and that each atom consists of electrons, protons, and neutrons. The presence of equal numbers of protons and electrons leaves an atom neutrally charged.



Helium

- The charge e on an electron $= -1.602 \times 10^{-19}\,C$
- The charge q on a proton $= +1.602 \times 10^{-19}\,C$
- The *law of conservation of charge* states that charge can neither be created nor destroyed, only transferred. Thus, the algebraic sum of the electric charges in a system does not change.
- *Charge quantization* is the principle that the charge of any object is an integer multiple of the elementary charge. $[0.5e, 3.7q, 1.2e$ *etc. charges don't exist*$]$

Problem 1

(i) How many electrons are there in $1C$ of charge?

(ii) How much charge is represented by these number of electrons?

$(a)\ 1.24 \times 10^{18}$
$(b)\ 2.46 \times 10^{19}$

Ans: $(i)\ \mathbf{6.25 \times 10^{18}}\ \boldsymbol{electrons}.$
$(ii)\ (a)\ -\ \mathbf{198.65\ mC}, (b)\ -\ \mathbf{3.941\ C}$

1. Calculate the amount of charge represented by 6.667 *billion* protons.

Answer: *1.0681×10^{-9}*

Electricity

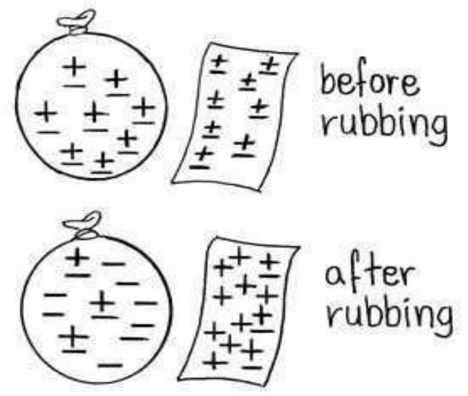
Electricity is a fundamental form of energy resulting from the movement or flow of electrically charged particles, typically electrons. Electricity can be divided into two categories based on how the charge behaves.

Static Electricity

Current Electricity

Static Electricity

Static electricity is electricity in which the charges remain at rest on the surface of a material. These “static charges” don’t go away unless they’re grounded or released. It induces because of the movement of the negative charges from one object to another.



Common Examples of Static Electricity

Balloon and Hair:

Balloon: When you rub a balloon on your hair, the balloon gains electrons from the hair.
Hair: The hair loses electrons to the balloon.
Result: The balloon becomes negatively charged (gains electrons), and the hair becomes positively charged (loses electrons).

Glass and Silk:

Glass: When you rub a glass rod with silk, the glass loses electrons to the silk.
Silk: The silk gains electrons from the glass.
Result: The glass rod becomes positively charged (loses electrons), and the silk becomes negatively charged (gains electrons).

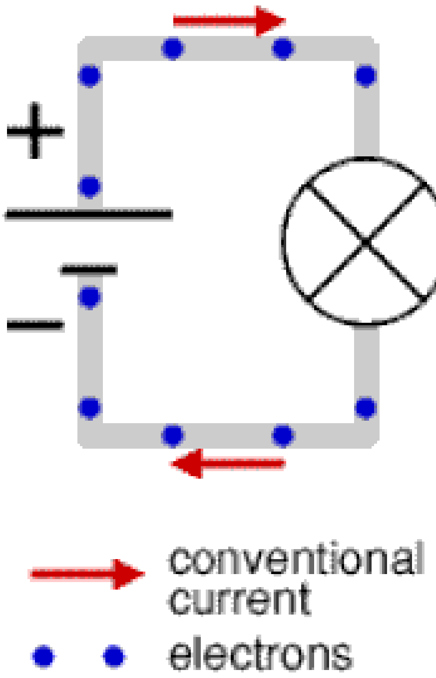
Rubber and Cat Fur:

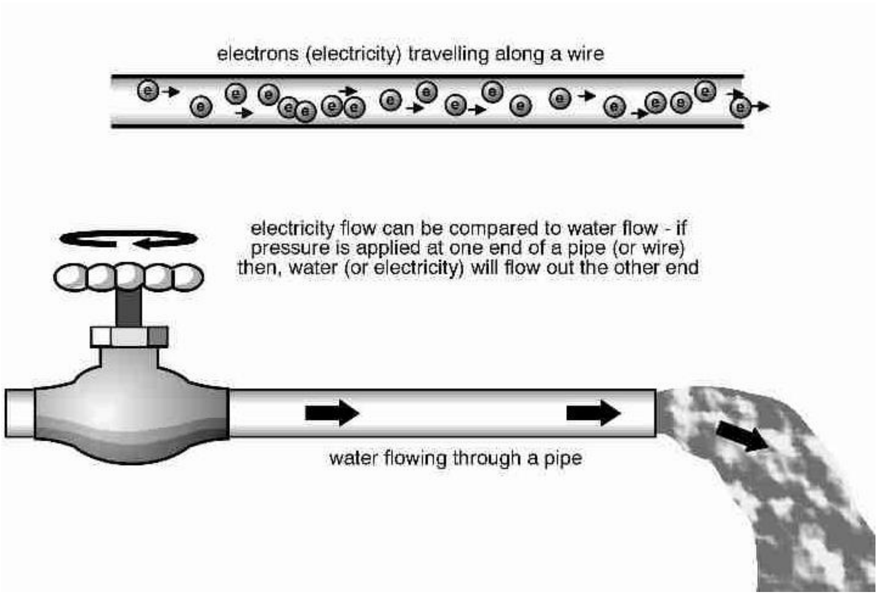
Rubber: When you rub a rubber rod on cat fur, the rubber gains electrons from the cat fur.
Cat Fur: The cat fur loses electrons to the rubber.
Result: The rubber becomes negatively charged (gains electrons), and the cat fur becomes positively charged (loses electrons)..

Wanna Know, which material prefers positive charges and which material prefers negative charges? See the appendix

Current Electricity

Current electricity is the electricity that is generated as a result of the movement of electrons. It can only form on materials with free electrons (Conductor). It develops only in the conductor.





Static vs Current Electricity

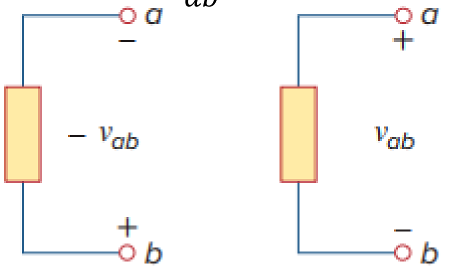
Static electricity	Current electricity
The electricity which is built up on the surface of the substance is known as static electricity.	The current electricity is because of the flow of electrons.
It induces because of the movement of the negative charges from one object to another	The current electricity is because of the continuous movement of the electrons.
The static electricity develops both in the conductor and insulator.	The current electricity develops only in the conductor.
Does not induce magnetic field.	It induces a magnetic field.
Lightning strikes, it develops by rubbing the balloons on hair, etc.	The current electricity is used for running the fan, light, T. V., etc.

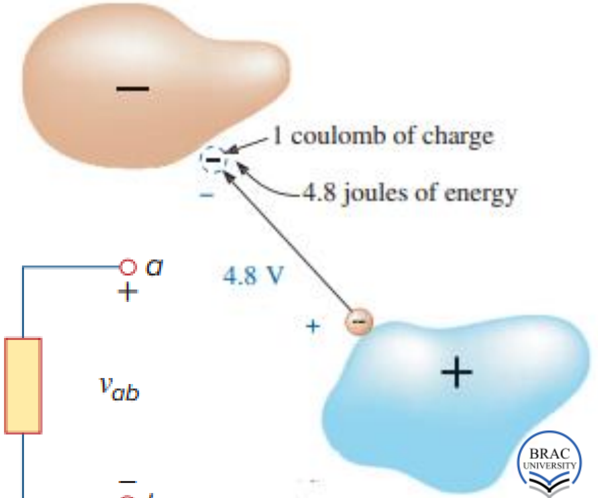
Voltage

Voltage is like the pressure in a water pipe, but for electricity. It's a measure of the "push" that makes electric charges move through a circuit.

- *Electric potential or potential difference or voltage* is the energy required **per unit** charge to move a charge from a reference point to another point of interest.
- The voltage V_{ab} between two points $\boldsymbol{a}$ and $\boldsymbol{b}$ in an electric circuit is the energy (or work) needed to move a positive unit charge from $\boldsymbol{a}$ to $\boldsymbol{b}$; mathematically,
- $V_{ab} = \frac{W}{Q} \quad \left(\frac{Joule}{Coulomb} = Volts\right)$
- The V_{ab} can be represented in two ways: (1) Point a is at a potential of V_{ab} volts higher than point b, or (2) the potential at point a with respect to point b is V_{ab}
- Logically, $V_{ab} = -V_{ba}$
- Is there anything like V_a or V_b?

Let's find on the next slide!





Absolute Potential

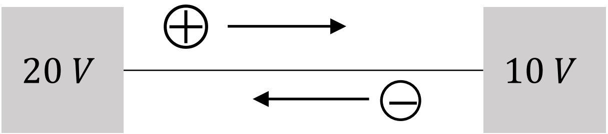
- The concept of *absolute potential* is based on the reality that *there is no such thing as absolute potential!* Let's reiterate the definition of potential difference.
- *Potential difference or voltage* is the energy required per unit charge to move a charge from a reference point to another point of interest.
- We consider the potential of the point of interest to be the absolute potential if the reference point is at an infinite distance. Where the potential is considered to be 0 at infinite. Mathematically,
- $V_a - V_\infty = V_a - 0 = V_a; \quad where, V_\infty = 0$

$$W = qV$$

- Work done to move a charge q from point A to point B is thus, $\boxed{W = q(V_B - V_A)}$
- Note that, W depends only on the potentials of the initial and final points, not on the path of the charge being moved

V_{Final} $V_{Initial}$

Sign Convention



- Consider the scenario where two bodies at potentials $10\,V$ and $20\,V$ are connected by a conducting wire. In which direction the charges will move?

$\Rightarrow$ Every system tends to reside in its lower energy state. For a $-\mathrm{ve}$ charge, the lower energy is toward the higher potential and vice versa for a $+\mathrm{ve}$ charge. So, $-\mathrm{ve}$ charges will move from the body at $10\,V$ toward the body at $20\,\mathrm{V}$ and $+\mathrm{ve}$ charges will move in the opposite direction until the two bodies are at equal potential of $15\,\mathrm{V}$. We say, this is the natural flow of charges. In this case, charges do the required work.

- If we are to move the charges in opposite to their natural direction, external work must be done. 'W' is $+ve$ if external work is done and $-ve$ if the work is done by the charge itself. This signing convention is in accordance with that of moving object vertically. Lifting an object in opposite to the gravitational force increases its potential energy ($+ve\ U$).

$$W = q(V_{Final} - V_{Initial})$$

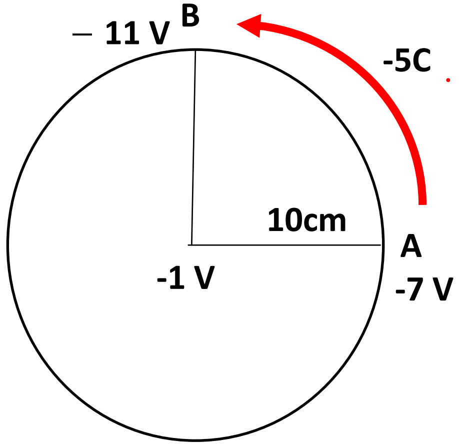
W: $+$ If external work is done; $-$ If work is done by charge itself

q: (for $+\,ve$ charge) $+$; $-$ (for $-\,ve$ charge)

Problem 2

(i) At point $\boldsymbol{A}$, there's a voltage of $V_A = 22\,V$. If a charge with $q = -2\,C$ moves from point A to another point B and while moving **it does a work** of $W = 10\,J$, what's the voltage of point $\boldsymbol{B}$?

(ii) How much work must be done to transport the $-5\,C$ charge from point $\boldsymbol{A}$ to point $\boldsymbol{B}$ around the circle depicted in the diagram?



Ans: (i) $\boldsymbol{V_B = 27\,V}$
(ii) $\boldsymbol{W = +20\,J}$

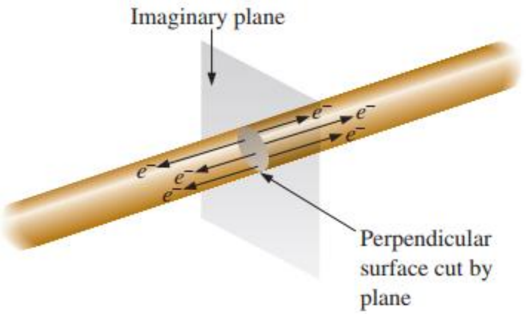
Problem 3

- Calculate the work that must be done in moving a $-6\ mC$ charge from point A to point B, where the potentials of points A and B are $20\ V$ and – $30\ V$ respectively. Is the work done by the system? $\boxed{\text{Ans: } \mathbf{0.3\ J}}$

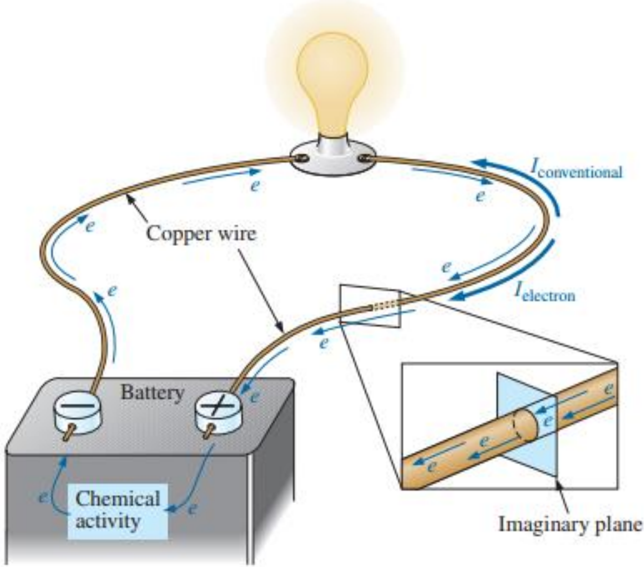
2.	If the potential difference between two points is $60\ V$, how much energy is expended to bring $8\ mC$ from one point to the other?	***Answer:*** *$\pm 0.48\ J$*
3.	How much charge passes through a radio battery of $9\ V$ if the energy expended is $72\ J$?	***Answer:*** *$\pm 8\ C$*
4.	To move charge q from point b to point a requires $25\ J$. Find the voltage drop v_{ab} if: (a) $q = 5\ C$, (b) $q = -10\ C$.	***Answer:*** *(a) 5V, (b) – 2.5 V*
5.	If $10\ J$ work is done on a – $2C$ charge in moving it from point A to point B, where $VB = 20\ V$, what is the potential of point A?	***Answer:*** *25 V*

Current

- *Electric current* is the time rate of change of charge, measured in amperes (A).
- Mathematically, the relationship between current i, charge q, and time t is,
- $i = \frac{dq}{dt}$ $(coulomb/\sec \equiv Ampere)$
- Free electrons generated at room temperature in a conductor are in constant motion in random directions, however, the net flow in any one direction is zero (current is zero).
- To make this electron flow do work for us, we need to give it a direction and be able to control its magnitude. This is accomplished by simply applying a voltage (driving force) across the wire to force the electrons to move toward the positive terminal of the battery.

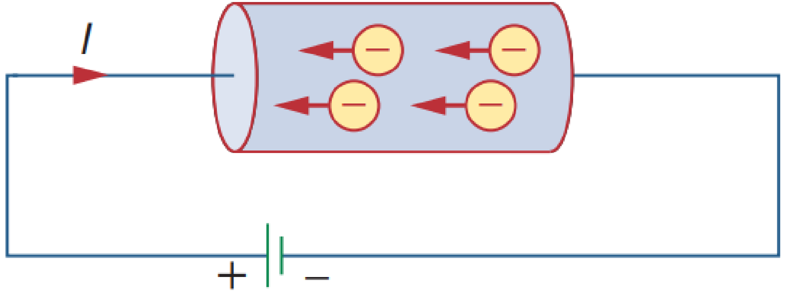


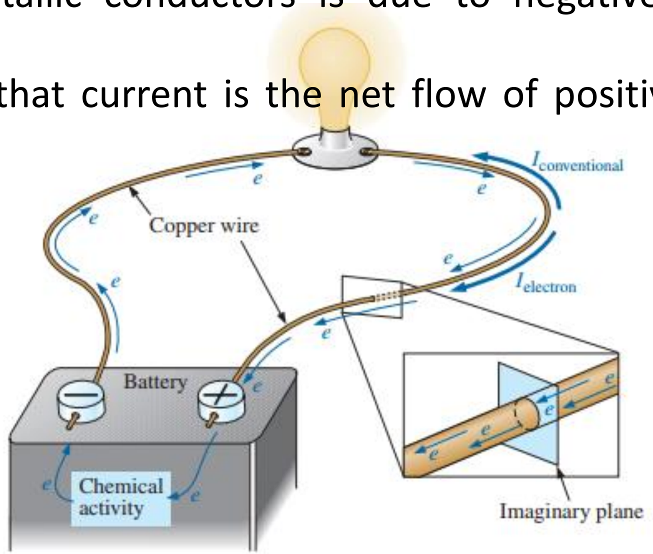
Isolated Cu wire with zero net flow of charge in any particular direction



Current: direction controversy

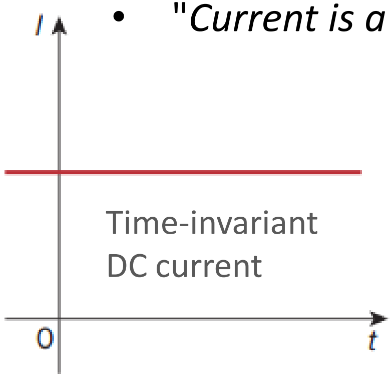
- There are two directions of charge flow: positive charges move in one direction while negative charges move in the opposite direction.
- The controversy in the direction of current is a result of an assumption made at the time electricity was discovered that the positive charge was the moving particle in metallic conductors although we know that current in metallic conductors is due to negatively charged electrons.
- We will follow the universally accepted convention that current is the net flow of positive charges.

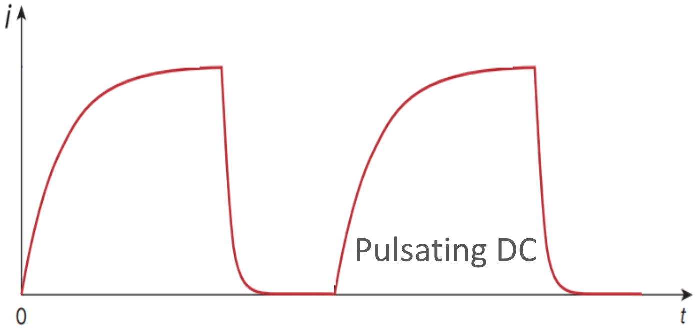


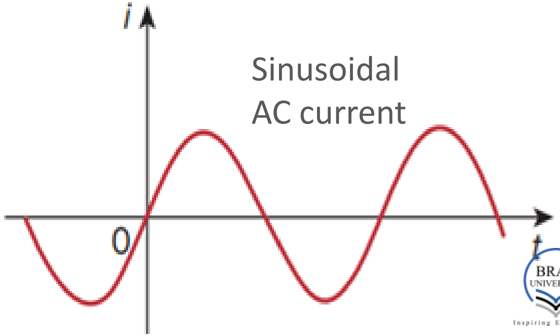


Current: types

- $i = \frac{dq}{dt}$ or $q = \int_{t_0}^{t} i\, dt$ suggests that current need not be a constant-valued function. It can vary in several ways. But there are two ways that current can flow. It can only flow in one direction or in both directions simultaneously.
- A *direct current (dc)* flows only in one direction and can be constant or time varying.
- An *alternating current (ac)* is a current that changes direction with respect to time. So, ac current is always time-varying. We will explore ac current more in detail later.
- "*Current is a scalar quantity in spite of having magnitude and direction*",----- think why!







Circuit Elements

- **Active element**
  - An *active element* is capable of generating energy.
  - In other words, an element is said to be active if it can add some gain (in terms of voltage or current) to a circuit.
  - Active elements can absorb energy if they are forced to do so by other active elements.
  - Examples: *Voltage/current sources, generators, transistors, operational amplifiers.*
- **Passive element**
  - *Passive elements* cannot supply energy. They can only consume/dissipate/store energy.
  - Examples: *Resistors, capacitors, inductors, transformers.*
  - Transformers change the voltage or current levels, but the power is unchanged. This is why transformers are passive element.

Power

- *Power* is the time rate of expending or absorbing energy, measured in watts (W).
- $p = \frac{dw}{dt}$, where p is power in watts (W), w is energy in joules (J), and t is time in seconds (s).

$$\Rightarrow p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = vi$$

$$\Rightarrow \boxed{p = vi} \quad [1\,Watt = 1\,Joule/\mathrm{sec}]\ [746\,Watt = 1\,horse - power\ (hp)]$$

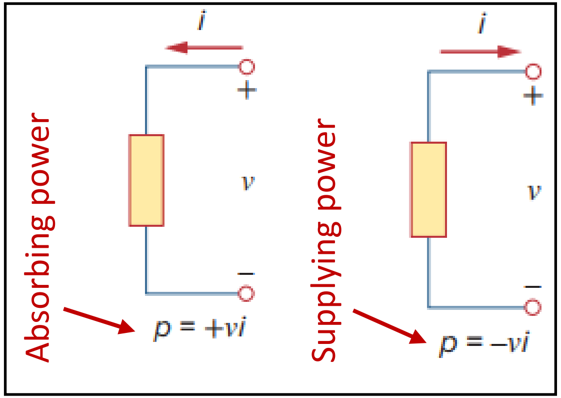
$$\Rightarrow w = \int_{t_0}^{t} p\,dt = \int_{t_0}^{t} vi\,dt$$

- The power p is a time-varying quantity and is called the *instantaneous power*.
- Power is a signed quantity as voltage and current are signed quantities. So, what does positive power and negative power mean? Let's find in the next slide.

Passive Sign Convention

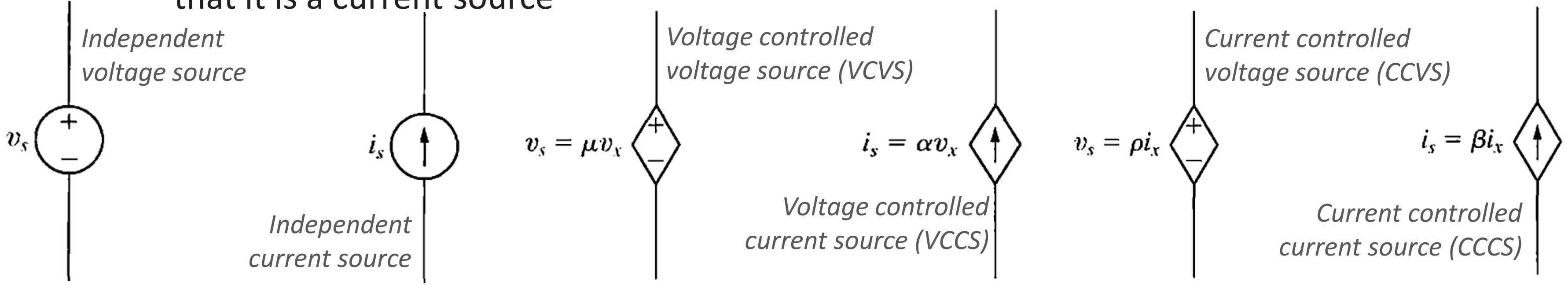
- The universal standard is that, if the power has a + sign, power is being delivered to or absorbed by the element. If, on the other hand, the power has a – sign, power is being supplied by the element. But how do we know when the power has a negative or a positive sign? *Current direction and voltage polarity determines the sign of power.*
- *Passive sign convention* is satisfied when the current enters through the positive terminal of an element and $p = +vi$. If the current enters through the negative terminal, $p = -vi$.
- In an electric circuit, $+Power\ absorbed = -Power\ supplied$
- The *law of conservation of energy* requires that, $\sum p = 0$

👉 **Determine first, by looking at the polarity of the voltage and the direction of the current, whether it is absorbing or supplying power. Then use $p = +vi\ or\ p = -vi$ accordingly.**



Electrical Sources

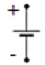
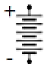
- An *electrical source* is a device that is capable of converting nonelectric energy to electric energy and vice versa. They can either deliver or absorb electric power, generally maintaining either voltage or current. *Ideal voltage sources* maintain a set voltage across their terminals regardless of current flow. *Ideal current sources* maintain a defined current via their terminals regardless of voltage.
- They can be further described as either independent sources or dependent sources.
- A *dependent source* or *controlled source* establishes a voltage or current whose value depends on the value of a voltage or current elsewhere in the circuit. The diamond shape of the symbol indicates that the device is a dependent source, and the arrow inside indicates that it is a current source



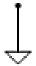
Circuit Symbols

- Basic Electrical and Electronic Schematic Symbols (Click here for more circuit symbols)

Power Supply Schematic Symbols

Schematic Symbol	Symbol Identification	Description of Symbol
	Single Cell	A single DC battery cell of 0.5V
	DC Battery Supply	A collection of single cells forming a DC battery supply
	DC Voltage Source	A constant DC voltage supply of a fixed value
	DC Current Source	A constant DC current supply of a fixed value
	Controlled Voltage Source	A dependent voltage source controlled by an external voltage or current
	Controlled Current Source	A dependent current source controlled by an external voltage or current
	AC Voltage Source	A sinusoidal voltage source or generator

Electrical Grounding Schematic Symbols

Schematic Symbol	Symbol Identification	Description of Symbol
	Earth Ground	Earth ground referencing a common zero potential point
	Digital Ground	A common digital logic circuit ground line

Resistor Schematic Symbols

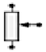
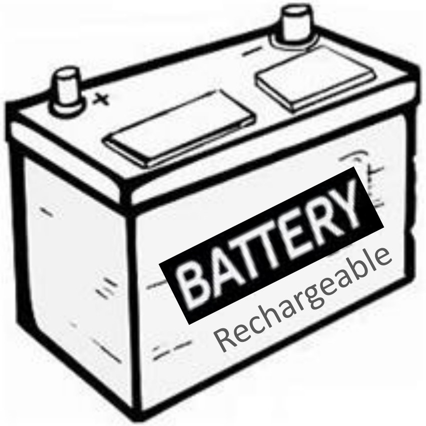
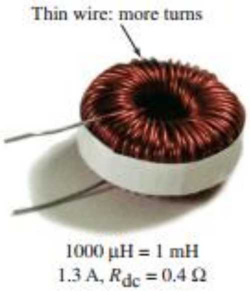
Schematic Symbol	Symbol Identification	Description of Symbol
	Fixed Resistor (IEEE Design)	A fixed value resistor whose resistive value is indicated next to its schematic symbol
	Fixed Resistor (IEC Design)	
	Potentiometer (IEEE Design)	Three terminal variable resistance whose resistive value is adjustable from zero to its maximum value
	Potentiometer (IEC Design)	
	Rheostat (IEEE Design)	Two terminal fully adjustable rheostat whose resistive value varies from zero to a maximum value
	Rheostat (IEC Design)	

Illustration of some basic circuit elements

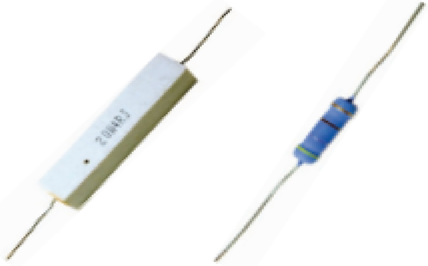


Function generator

Capacitor



Inductor

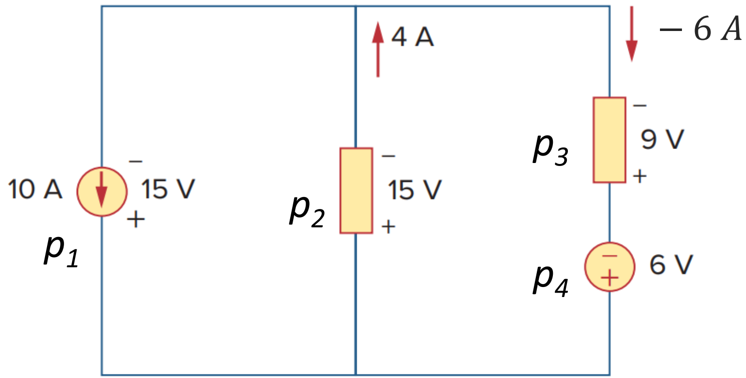


Fixed resistors

Variable resistors

Example 1

- Find power of each element in the network. Specify for each if it is supplying or absorbing.



Element 1

Current is leaving the + terminal of the voltage. So, according to the passive sign convention, $E1$ is supplying power.

$$p_1 = -15 \times 10 = -150\ (W)$$

Element 2

Current is entering the + terminal of the voltage. So, according to the passive sign convention, $E2$ is absorbing power.

$$p_2 = +15 \times 4 = +60\ (W)$$

Element 3

Current (the arrow) is leaving the + terminal of the voltage. So, according to the passive sign convention, it seems that $E2$ is supplying power.

$$p_3 = -9 \times (-6) = +54\ (W)$$

However, as the final form of the calculation is $+ve$, it is actually absorbing.

Element 4

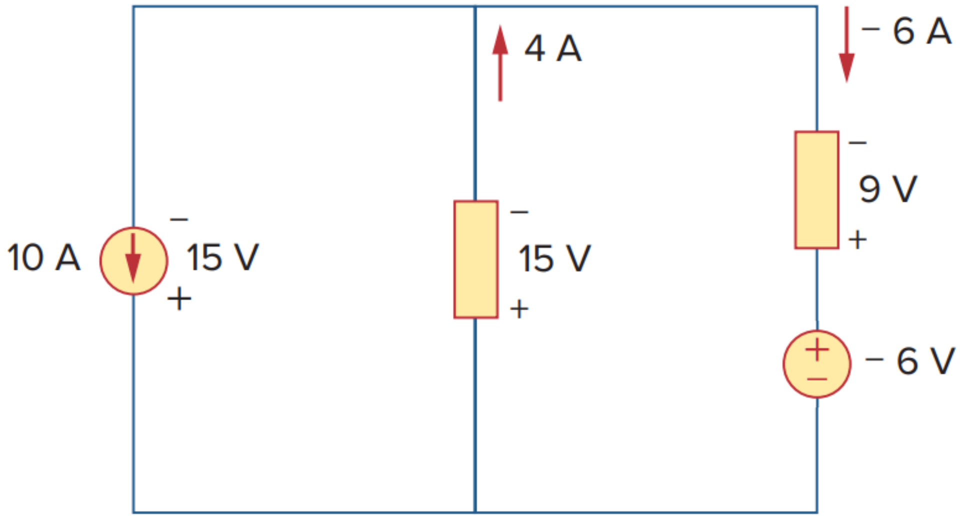
Current is leaving the + terminal of the voltage. So, according to the passive sign convention, it seems that $E2$ is supplying power.

$$p_4 = +6 \times 6 = +36\ (W)$$

However, as the final form of the calculation is $+ve$, it is actually absorbing.

Problem 4

- Find power of each element in the network. Specify for each if it is supplying or absorbing.

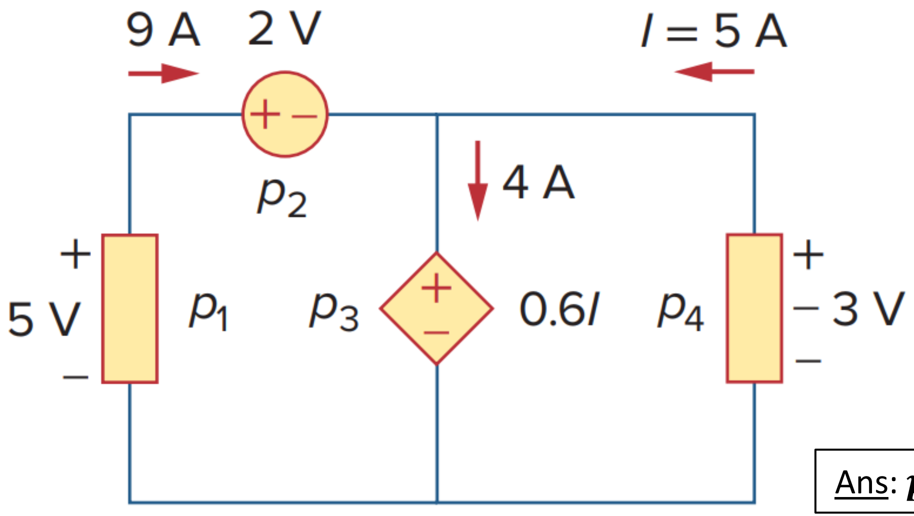


Ans: $p_1 = -150\ W, p_2 = 60\ W, p_3 = 54\ W, p_4 = 36\ W$

Problem 5

- Find power of each element in the network. Specify for each if it is supplying or absorbing.

.

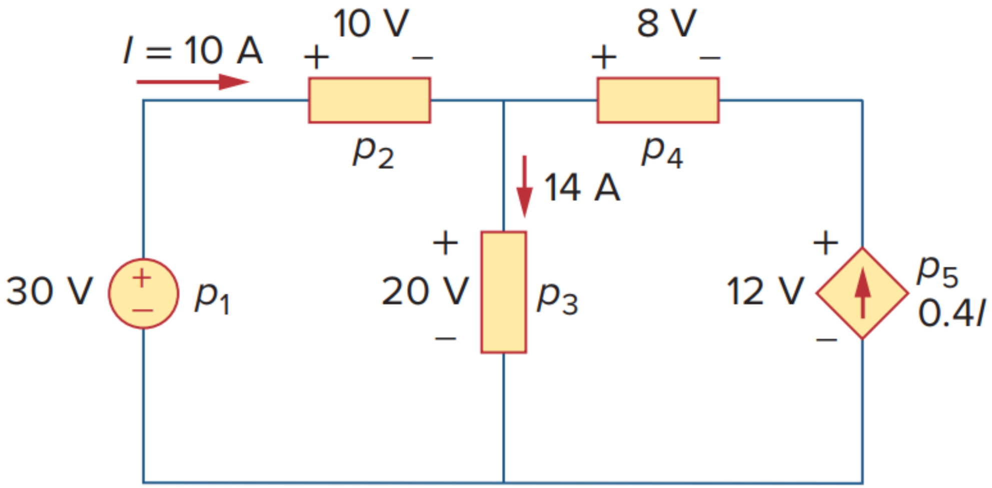


Ans: $p_1 = -45\ W, p_2 = 18\ W, p_3 = 12\ W, p_4 = 15\ W$

Problem 6

- Find power of each element in the network. Specify for each if it is supplying or absorbing.

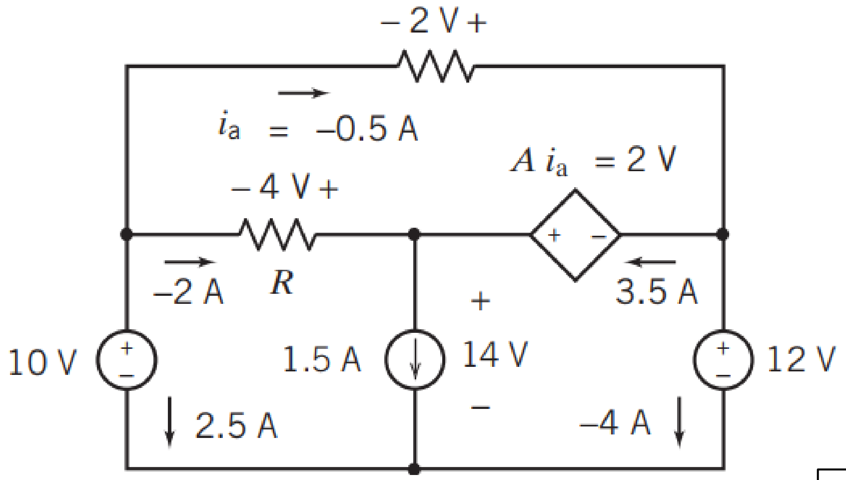
.



Ans: $\boldsymbol{p_1 = -300\,W, p_2 = 100\,W, p_3 = 280\,W, p_4 = -32\,W, p_5 = -48\,W}$

Problem 7

- Determine the value of the resistance R and the gain A of the CCVS.
- Find power of each element in the network. Specify for each if it is supplying or absorbing.



Solution:

$$R = \frac{-4}{-2} = 2\ (\Omega) \text{ and } A = \frac{2}{i_a} = \frac{2}{-0.5} = -4\ \left(\frac{V}{A}\right)$$

The rest of the steps are same as previous examples

Ans: $\mathbf{4,\ -4\ V/A,\ 25\ W,\ 8\ W,\ 21\ W,\ 1\ W,\ -7\ W, -48\ W}$

Calculation of Electric Bill

What is a kW and a kWh?

A “watt” is the unit used to measure quantities of power and is named after the Scottish inventor and engineer James Watt (1736-1819). A kilowatt, or kW, is equal to a thousand watts. So the number of kW is the amount of power an electrical device uses in order to run, and a kilowatt-hour (kWh) is the amount of energy that an appliance uses every hour. For example, if your electric radiator is rated at 3 kW and is left on for an hour, it would use 3 kWh of electricity.

More importantly, a kWh is the unit that electricity suppliers use to bill you for the electricity you use. They do this by either reading your usage for you, or by having you send them the reading from your meter. Usually, you are given a unit charge for your electricity; this multiplied by the number of kWh you use gives you the cost of the electricity on your bill.

How do you calculate the number of kWh from watts?

If you want to know how many kWh an appliance uses and already know how many watts it uses, the calculation is pretty straightforward.

First, you need to convert the number of watts into kW. To do that, you divide the number of watts by 1,000. So 100 W is 0.1 kW, 60 W is 0.06 kW, and 1500 W is 1.5 kW.

To get the number of kWh, you just multiply the number of kW by the number of hours the appliance is used.

For example, a device rated at 1,500 W that’s on for 2.5 hours:

1500 ÷ 1000 = 1.5. That’s 1.5 kW. 1.5 x 2.5 = 3.75. So, a 1,500 W appliance that’s on for 2.5 hours uses 3.75 kWh.

https://www.inspirecleanenergy.com/blog/sustainable-living/how-to-calculate-kwh-kilowatt-hours

Problems

1. A utility company charges 8.2 cents/kWh. If a consumer operates a 60-W light bulb continuously for one day, how much is the consumer charged?

2. A 1.2-kW toaster takes roughly 4 minutes to heat four slices of bread. Find the cost of operating the toaster twice per day for 2 weeks (14 days). Assume energy costs 9 cents/kWh.

Practice Problems

- Additional recommended practice problems: here
- Other suggested problems from the textbook: here

Fundamentals of electric circuits (5th edition) – C.K. Alexander and M.N.O. Sadiku

Chapter 1

Example – **1-7.**

Practice Problem – **1-6.**

Exercise – **1-7, 11, 13, 15, 16, 18, 21-25.**

Appendix

Triboelectric Series
The triboelectric series ranks materials from most likely to give up electrons (becoming positively charged) to most likely to accept electrons (becoming negatively charged). Here is a simplified version of the triboelectric series:

More likely to lose electrons (become positively charged):
Glass
Human hair
Nylon
Wool
Fur

Neutral tendency (midpoint):
Cotton

More likely to gain electrons (become negatively charged):
Silk
Rubber
Polyester
Plastic (PVC)
Teflon